FRESHERS ACEDAMIC ROADMAP STRATEGY GUIDE

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(DEEMED TO BE UNIVERSITY)

ABSTRACT

The Freshers Academic Roadmap is a web-based platform designed to help first-year engineering students plan and track their academic journey. The system provides branch-specific learning roadmaps, progress tracking, gamification with XP and badges, exam countdown management, and curated study resources. The application is developed using HTML, CSS, and JavaScript for the frontend, while Data Structures and Algorithms concepts such as arrays, stack, queue, searching, and sorting are demonstrated through a Java console program. This project provides a structured and engaging approach to academic planning for freshers.

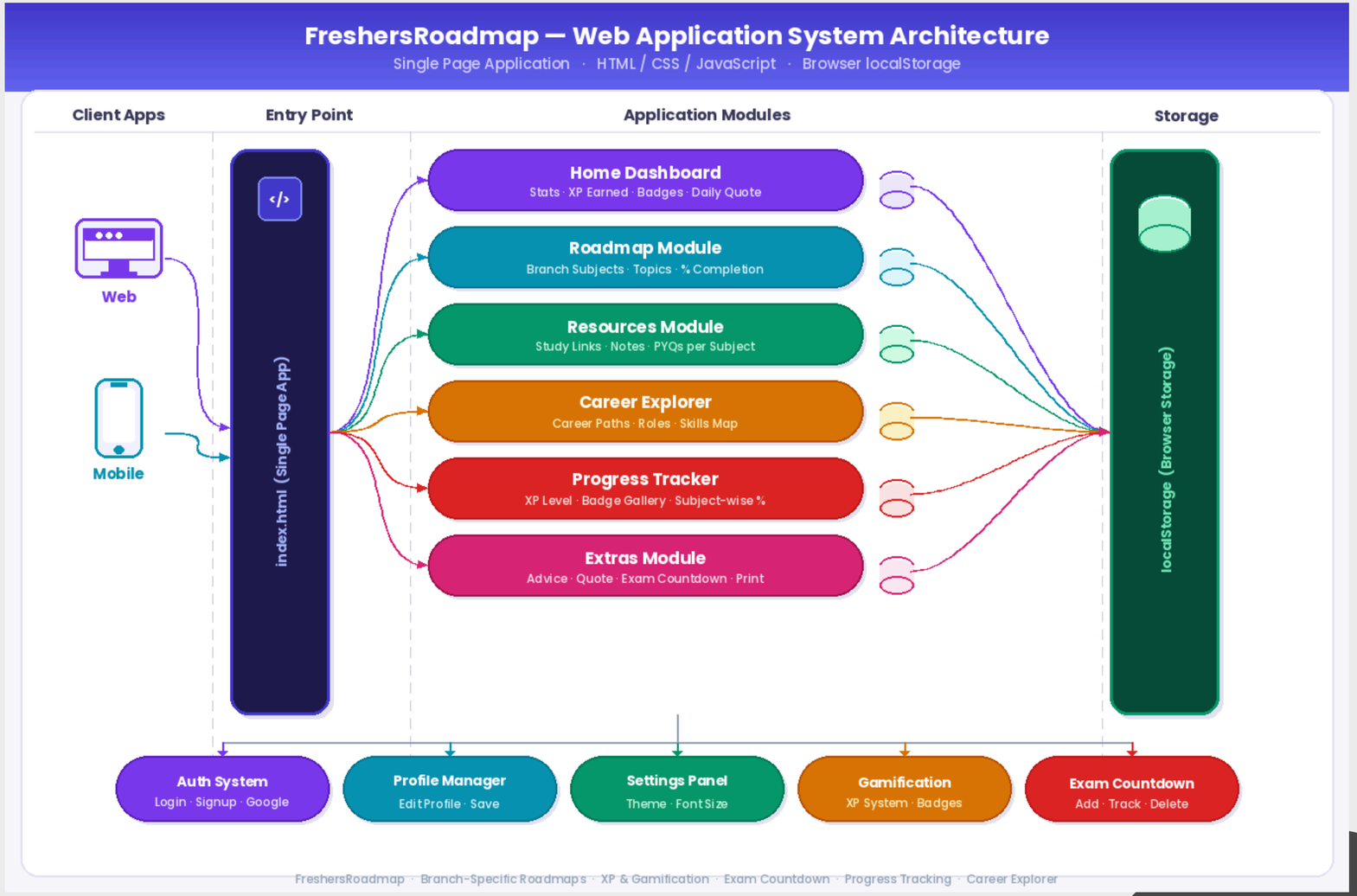
EXISTING SYSTEM

- No structured academic roadmap for first-year students
- Students depend on random YouTube videos or websites
- College portals only show attendance and marks
- No system to track topic-wise academic progress
- No motivation system like achievements or rewards
- No centralized platform for study resources and career guidance
- Students lack awareness about skills required for future careers
- Academic planning mostly done manually without tracking tools

PROPOSED SYSTEM

- Web-based academic guidance platform for engineering students
- Branch-specific semester-wise roadmap with subject topics
- Interactive topic completion tracking system
- Gamification system with XP points and achievement badges
- Exam countdown manager for planning upcoming exams
- Study resource links for each subject topic
- Career path explorer for different technical roles
- User-friendly interface built with HTML, CSS and JavaScript
- Integration of DSA concepts using Java console program

SYSTEM ARCHITECTURE



PROJECT MODULES

User Authentication Module — User registration, login, profile management and secure account handling using browser localStorage.

Academic Roadmap Module — Displays branch-wise subjects with topic-level breakdown for structured learning guidance.

Progress Tracking Module — Tracks completed topics and shows real-time progress percentage for each subject.

Gamification Module — Awards XP points, levels and achievement badges to motivate students to complete topics.

Exam Countdown Module — Allows students to add exam dates and displays live countdown with urgency alerts.

PROJECT MODULES

Study Resources Module — Provides curated learning resources such as NPTEL, MIT OCW, YouTube and GeeksforGeeks.

Career Path Explorer Module — Shows engineering career paths with required skills and job roles.

DSA Demonstration Module — Implements Stack, Queue, Linear Search and Bubble Sort using a Java console program.

Theme & Data Persistence Module — Dark/light mode support with user data stored using browser localStorage.

OBJECTIVES

- To develop a web-based academic planning platform for freshers
- To provide structured subject roadmaps for engineering branches
- To help students track topic-wise learning progress
- To motivate students through gamification elements
- To assist students in planning exams effectively
- To provide curated study resources for each subject
- To introduce career guidance for engineering students
- To demonstrate practical application of DSA concepts using Java

CO’S ATTAINMENT- FWD

CO1 —

- Semantic HTML tags used for webpage structure
- CSS variables, selectors and box model applied for styling and spacing

CO2 —

- Login & Signup forms used with email validation and required fields
- Flexbox and Grid used for dashboard and card layout
- CSS animations and media queries used for responsive design

CO3 —

- Functions, conditions and loops used for roadmap logic and XP calculation
- Arrays and objects (BRANCH_DATA, CAREER_DATA) used to store branch and topic data
- Array methods map, filter, for Each used to dynamically render cards

- **CO4** —DOM manipulation used to build subject cards and topic lists dynamically
- Event handling used for topic toggle, login, exam add and badge unlock
- localStorage used to save user progress and exam data

- **CO5** —Form validation logic used for email check and inline error display
- setInterval and date logic used for exam countdown timer
- Entire project deployed as a single static HTML file with no backend

CO’S ATTAINMENT- DSA

CO1 — Searching and Sorting Algorithms
- Linear Search used to locate specific subjects or topics
- Bubble Sort used to arrange data in order

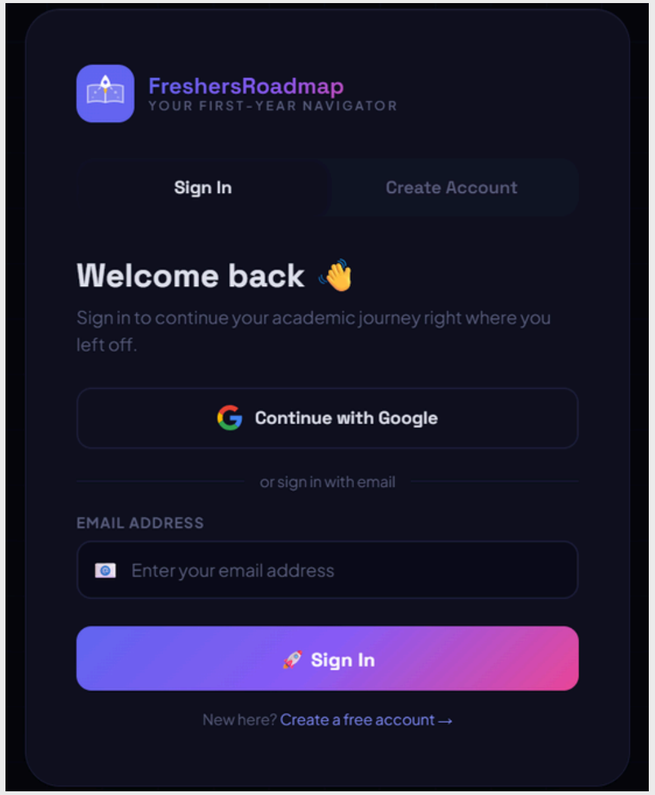
CO2 — List and Abstract Data Types
- Arrays used to store subjects and topic records
- List operations such as insertion and traversal demonstrated

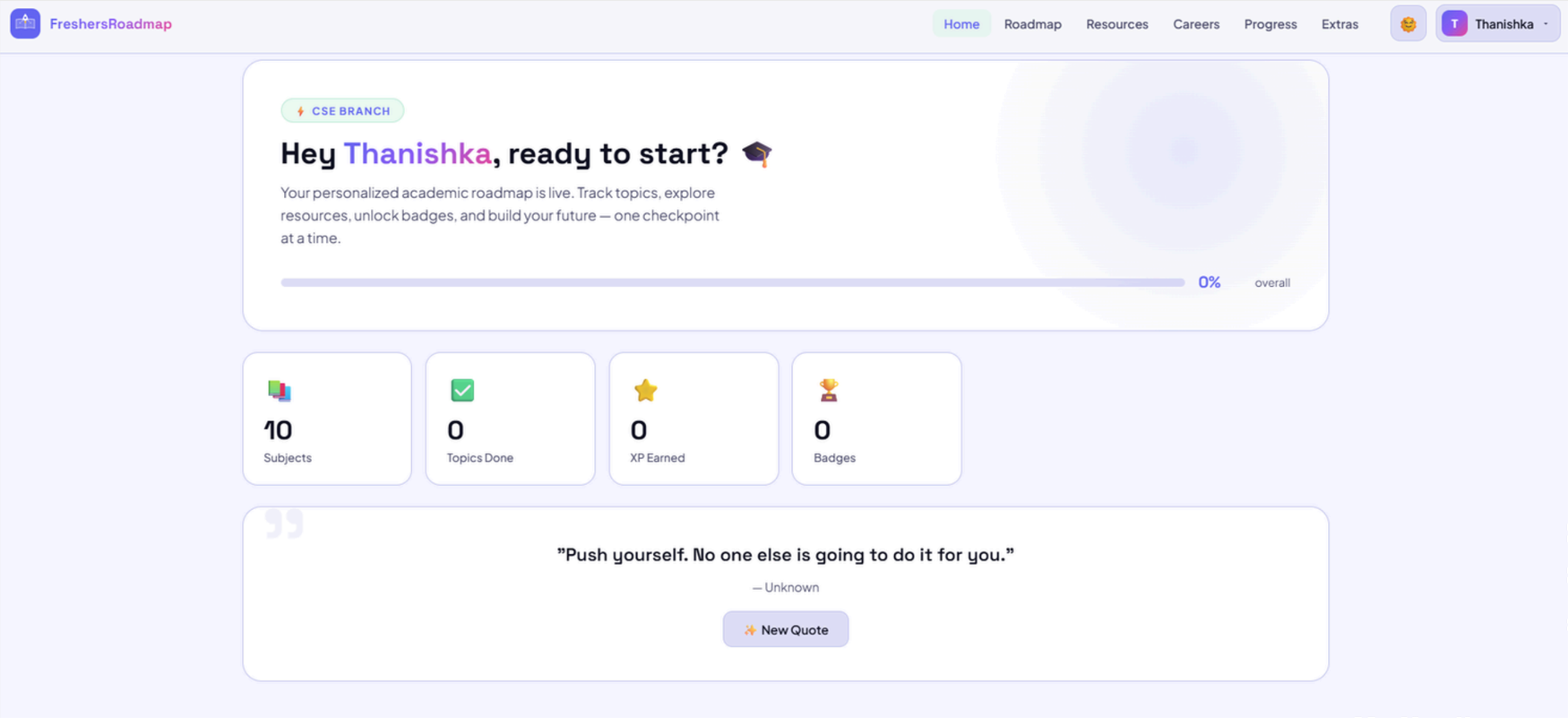
CO3 — Stack and Queue Concepts
- Stack implemented using push and pop operations
- Queue implemented using enqueue and dequeue operations

CO4 — Data Handling and Storage
- Data structures used for efficient management of records
- Structured data representation using arrays and objects

CO5 — Application of DSA Concepts
- Java program created to demonstrate stack, queue, searching and sorting
- Program output displayed through console during project demonstration

SCREENSHOTS- FWD





📍 Academic Roadmap

Check off topics as you complete them. Sem 2 unlocks after 80% of Sem 1.

Semester 1 | Semester 2 🔒

Engineering Mathematics I

40%

MA101

Calculus | Algebra | Problem Solving

- [x] ~~Matrices & Determinants~~
- [x] ~~Differential Calculus~~
- [ ] Integral Calculus
- [ ] Series Expansion
- [ ] Vector Calculus & Applications

📚 Study Resources →

Programming in C

0%

CS101

Coding | Debugging | Logic Building

- [ ] Variables, Data Types & Operators
- [ ] Control Flow — if/else/loops
- [ ] Functions & Recursion
- [ ] Arrays, Strings & Pointers
- [ ] File I/O & Structures

📚 Study Resources →

Engineering Physics

0%

PH101

Physics | Analytical Thinking

- [ ] Quantum Mechanics Basics
- [ ] Wave Optics — Interference & Diffraction
- [ ] Semiconductor Physics
- [ ] Lasers & Fiber Optics
- [ ] Nanotechnology Intro

📚 Study Resources →

Engineering Chemistry

0%

CY101

Chemistry | Environmental Awareness

- [ ] Electrochemistry & Cells
- [ ] Corrosion & Protection
- [ ] Polymers & Applications
- [ ] Fuels & Combustion
- [ ] Water Technology & Treatment

📚 Study Resources →

Engineering Drawing

0%

EG101

Design | Visualization | CAD

- [ ] Orthographic Projection
- [ ] Isometric & Sectional Views
- [ ] Auxiliary & Development Views
- [ ] AutoCAD Basics
- [ ] 3D Solid Modeling

📚 Study Resources →

📖 Study Resources

Real, working links — videos, notes, practice, tools. All verified.

🎥 VIDEO LECTURES

NPTEL — IIT Courses (Free)

Official free video lectures from IIT/IISc professors covering CS, Maths, Physics. Certificate courses available.

📌 All Subjects

Open Resource →

💻 CODING PRACTICE

GeeksforGeeks

Largest CS portal — DSA, algorithms, system design, company interview questions and theory for all CS subjects.

📌 Programming / CS

Open Resource →

💻 CODING PRACTICE

LeetCode

Industry-standard platform for coding interviews. 2500+ problems with company-wise tagging. Start early, win big.

📌 Algorithms & DSA

Open Resource →

🎥 VIDEO LECTURES

Abdul Bari — DSA (YouTube)

The best Data Structures & Algorithms playlist on YouTube. Clear, detailed, with full examples.

📌 Data Structures

Open Resource →

🎥 VIDEO LECTURES

Jenny's Lectures CS (YouTube)

Beginner-friendly C programming and DSA lectures. Step-by-step and perfect for first-year students.

📌 Programming / DSA

Open Resource →

🎥 VIDEO LECTURES

Gate Smashers (YouTube)

In-depth CS core subjects — TOC, OS, CN, DBMS, Digital Logic. Essential for exams.

📌 CS Core Subjects

Open Resource →

🔧 TOOLS

OnlineGDB — C/C++ Compiler

Free browser-based C/C++ IDE. Write, compile, debug online without installing anything.

📌 Programming Lab

Open Resource →

🔧 TOOLS

Wolfram Alpha

World's most powerful computational engine. Solve any maths problem with step-by-step solutions.

📌 Mathematics

Open Resource →

📘 TEXTBOOKS

Archive.org — Free Books

Free digital library with engineering textbooks — Kreyszig, Morris Mano, Tanenbaum and more.

📌 All Textbooks

Open Resource →

💻 CODING PRACTICE

HackerRank

Practice C, Python, Java with structured challenges.

Your Progress

XP, levels, badges, and subject-wise tracking.

Experience Points

Level 1 — Fresher

40 XP

460 XP to next level

Achievement Badges

First Step
Completed your 1st topic
UNLOCKED ✓

Bookworm
Completed 10 topics

On Fire
Completed 20 topics

Goal Setter
Completed 5 topics

Bright Mind
Completed 30 topics

Rising Star
Sem 1 – 50% done

Semester Champion
Sem 1 – 100% complete

Scholar
Completed 40 topics

Career Path Explorer

Where your branch can take you — roles, skills, and roadmaps.

Software Developer

Build applications, web services, APIs and platforms. Highest demand across every sector globally.

Java/Python/JS · DSA · System Design · Git/GitHub

AI / ML Engineer

Design and train intelligent models. Fastest growing and highest compensated field in tech.

Python · Maths/Stats · TensorFlow/PyTorch · Data Pipelines

Cloud / DevOps Engineer

Architect scalable infrastructure and automate deployment pipelines. Every company is moving to cloud.

AWS/Azure/GCP · Linux/Bash · Docker/Kubernetes · CI/CD

Cybersecurity Analyst

Protect systems from cyber threats. Critical in banking, defense, healthcare and every digital sector.

Networking · Cryptography · Ethical Hacking · SIEM Tools

Data Scientist

Extract insights from massive datasets to drive business decisions. Combines stats, programming and domain knowledge.

Python/R · SQL · Statistics · Data Viz

Full Stack Developer

Build complete web products — pixel-perfect frontends to robust backends and databases.

React/Vue · Node.js · SQL/MongoDB · REST/GraphQL

Exam Countdown

+ Add / Manage Exams

No exams added yet

Add your exam dates to start the countdown.
You'll see days remaining, urgency levels & more.

+ Add Your First Exam

Advice from Seniors

KR Karthikeyan R. ✓
Final Year, CSE
NIT Trichy
9.1 CGPA | Google SWE Intern

"Start DSA from semester 1, not after. I spent my entire first semester thinking I had time. Array problems alone took two months to get comfortable with. LeetCode Easy → Medium → Hard is the only real path. The earlier you start, the more relaxed your placement year will be."

Helpful 247 found this helpful

MS Meghana Subramaniam ✓
3rd Year, CSE
BITS Pilani, Goa
8.7 CGPA | Microsoft SDE Intern

"Don't bunk labs. I got my internship because I could demo an actual embedded system I built in a first-year electronics lab. That single project carried my resume more than my coursework. Labs are where theory becomes real — treat them seriously."

Helpful 189 found this helpful

AM Arjun Mehta ✓
Final Year, ECE
IIT Bombay
8.4 CGPA | Qualcomm — VLSI Engineer

"For ECE students — learn Verilog and C in semester 1 itself. Everyone regrets not doing it earlier. The gap between understanding digital theory and writing actual RTL is huge and takes months. Start a simple UART or ALU project as soon as you learn flip-flops."

Helpful 178 found this helpful

PR Priya Raghunathan ✓
3rd Year, AI&DS
PSG Tech, Coimbatore
9.3 CGPA | Research Intern, IISc Bangalore

"Mathematics is not your enemy. Linear algebra and probability from first year are literally the backbone of every ML model. I spent extra time with my maths professor in semester 1 and it paid off immensely by third year. Build a strong foundation — it compounds."

Helpful 214 found this helpful

RN Rohit Nambiar ✓
Final Year, IT
VIT Vellore
8.9 CGPA | Flipkart — SDE 1

"Build something visible in first year. Even a simple weather app or a calculator on GitHub matters. The habit of shipping completed projects is rarer than you think. By final year, students with 10 GitHub projects have a massive edge over those who just scored marks."

Helpful 156 found this helpful

AK Ananya Krishnan ✓
3rd Year, MECH
College of Engineering, Guindy (CEG)
8.2 CGPA | DRDO Project Intern

"Join a technical club in the first month — SAE, robotics, BAJA, anything. I joined CEG's SAE chapter in week two. The real-world projects, tight deadlines, and mentorship from seniors gave me more practical skills than two years of classroom theory combined."

Helpful 143 found this helpful

T Thanishka
CSE • Sem 1

- Edit Profile
- Settings
- My Progress
- Log Out
- Delete Account

SCREENSHOTS- DSA

```java
public class FreshersAcademicRoadmap {
    // ============================================================
    //  STACK ADT implemented using Singly Linked List
    //  Supports: push, pop, peek — O(1) for all
    // ============================================================
    static class StackNode {
        String data;
        StackNode next;
        StackNode(String d) { this.data = d; }
    }

    static class StudyStack {
        StackNode top;

        // PUSH: insert at top
        void push(String data) {
            StackNode node = new StackNode(data);
            node.next = top;
            top = node;
        }

        // POP: remove from top
        String pop() {
            if (isEmpty()) return null;
            String data = top.data;
            top = top.next;
            return data;
        }
```

```java
public class FreshersAcademicRoadmap {
    // ============================================================
    //  QUEUE ADT implemented using Singly Linked List
    //  FIFO: enqueue at rear, dequeue from front
    // ============================================================
    static class QueueNode {
        String data;
        QueueNode next;
        QueueNode(String d) { this.data = d; }
    }

    static class TaskQueue {
        QueueNode front, rear;

        // ENQUEUE: add at rear
        void enqueue(String data) {
            QueueNode node = new QueueNode(data);
            if (rear == null) { front = rear = node; return; }
            rear.next = node;
            rear = node;
        }

        // DEQUEUE: remove from front
        String dequeue() {
            if (isEmpty()) return null;
            String data = front.data;
            front = front.next;
            if (front == null) rear = null;
            return data;
```

```java
public class FreshersAcademicRoadmap {

    // ============================================================
    //  SEARCHING ALGORITHMS
    // ============================================================

    /**
     * LINEAR SEARCH - O(n)
     * Scans each element sequentially until target is found.
     * Works on unsorted data.
     */
    static int linearSearch(String[] arr, String target) {
        for (int i = 0; i < arr.length; i++)
            if (arr[i].toLowerCase().equals(target)) return i;
        return -1;
    }

    /**
     * BINARY SEARCH - O(log n)
     * Repeatedly halves the search space.
     * Requires sorted input.
     */
    static int binarySearch(String[] arr, String target) {
        int lo = 0, hi = arr.length - 1;
        while (lo <= hi) {
            int mid = (lo + hi) / 2;
            int cmp = arr[mid].toLowerCase().compareTo(target);
            if (cmp == 0) return mid;
            else if (cmp < 0) lo = mid + 1;
            else hi = mid - 1;
```

```java
public class FreshersAcademicRoadmap {

    /**
     * INSERTION SORT - O(n²)
     * Builds sorted portion one element at a time by inserting into correct position.
     */
    static void insertionSortArr(String[] arr) {
        int n = arr.length;
        for (int i = 1; i < n; i++) {
            String key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j].compareToIgnoreCase(key) > 0) {
                arr[j + 1] = arr[j]; j--;
            }
            arr[j + 1] = key;
        }
    }

    /**
     * MERGE SORT - O(n log n)
     * Divide-and-conquer: splits array, sorts halves, merges them.
     */
    static void mergeSort(String[] arr, int left, int right) {
        if (left < right) {
            int mid = (left + right) / 2;
            mergeSort(arr, left, mid);
            mergeSort(arr, mid + 1, right);
            merge(arr, left, mid, right);
        }
    }
```

CONCLUSION

- Successfully developed a web-based academic roadmap platform for freshers
- Implemented interactive features such as roadmap tracking, gamification and exam planning
- Built the frontend using HTML, CSS and JavaScript without external frameworks
- Demonstrated DSA concepts including stack, queue, searching and sorting using Java
- Provided a structured solution to help students plan and manage their academic journey
- The project shows how web development and data structures can be combined to solve real-world problems

GEO-TAG PHOTOS

THANK YOU